

Dipayan Dhar

COMPUTER ENGINEERING STUDENT

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EDUCATION

Memorial University of Newfoundland

St. John's, NL

Bachelor of Engineering (Co-op) | 2nd Year Computer Engineering | GPA: 4.00/4.00 (93.1%) September 2024 – Present

- Awards/Scholarships: **2024** – Memorial University Entrance Scholarship, Centenary of Responsible Government Scholarship, PEGNL Past President's Scholarship; **2025** – Hebron Diversity Award, Cenovus Energy Scholarship in Engineering, Verafin Inc. Computer Engineering Scholarship
- Dean's List (Fall 2025)

EXPERIENCE

C-CORE

St. John's, NL

MUNStars Satellite Software/Electrical Engineer

January 2026 – April 2026

- Designed a priority-based onboard telemetry and data handling system for the MUNStar-1 CubeSat, implementing structured file-based storage and retrieval for satellite telemetry data.
- Developed embedded C software for telemetry collection, request handling, and file management within a modular Mission Control Subsystem; validated system behavior on a ZedBoard FPGA platform through unit and system-level testing.
- Defined and encoded telemetry data structures using Protocol Buffers (Protobuf) to standardize communication between spacecraft subsystems and ground-station telemetry systems.

MUN Faculty of Engineering and Applied Science

St. John's, NL

Engineering Assistant

May 2025 – August 2025

- Conducted an open water propeller experiment on an Office of Naval Research Tumblehome (ONRT) propeller, running trials on 58m long towing tank
- Developed a Python function using the nptdms library to extract raw trial data by averaging signal values over the test duration
- Implemented Regex with a file naming convention to batch-process 184 trials automatically, generating a structured spreadsheet of test cases and results, saving hours of manual processing

PROJECTS

Paradigm Engineering | Software/Firmware Team Member

September 2025 – Present

- Collaborated with electrical, software, and mechanical subteams to develop autonomous control systems for a racing kart competing in the Autonomous Karting Series competition (2026).
- Developed embedded C firmware for an STM32G4 microcontroller using timer interrupts and PWM signal capture for steering and braking systems; enabled PWM communication with an NVIDIA Jetson for real-time motor control, validating system behavior using Pixhawk 6C Flight Controller.

Personal Website/Portfolio

August 2025 – April 2026

- Developed a personal portfolio website using HTML, CSS, and JavaScript, featuring interactive navigation and project showcases.
- Implemented dynamic UI components including a Swiper carousel and responsive layouts using Flexbox and CSS Grid.

Remote-Controlled Rescue Car

January 2025 – April 2025

- Designed and built a remote-controlled rescue vehicle using an Arduino microcontroller, integrating multiple hardware components with Python.
- Developed obstacle avoidance logic using ultrasonic sensor input and created a PyGame interface for vehicle control and sensor visualization.

PLA Filament Recycling Project

September 2024 – December 2024

- Designed a hand-powered grinder and remoulding system to recycle scrap PLA from the MUN Student Design Hub.
- Developed 3D models of grinder blades and enclosure using Onshape and 3D printed a prototype of the 7-blade assembly.

SKILLS & INTERESTS

- **Programming Languages:** C, C++, Python
- **Hardware/Embedded Systems:** STM32, Arduino, Raspberry Pi, FreeRTOS, STM32CubeIDE, PSpice, Simulink
- **Tools & Frameworks:** Git, Linux, Make, CMake, Protocol Buffers, Claude Code
- **Interests:** Embedded Systems, Robotics, Badminton, Chess, Guitar, Volunteering